



SCHOOL OF COMPUTER TECHNOLOGY

AASD 4007
Design Thinking for A.I. Solutions
Session 1

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Session 1



- Introduction to design thinking
- User research
- Introduction to MURAL
- IBM Design Thinking Practitioner (2hr Course)

Introduction to design thinking



Design thinking:

the Loop

Understand the present and envision the future in a continuous cycle of observing, reflecting, and making.



Observe

Immerse yourself in the real world with design research.



Reflect

Come together and look within to synchronize your movements, synthesize what you've learned, and share your "aha" moments with each other.

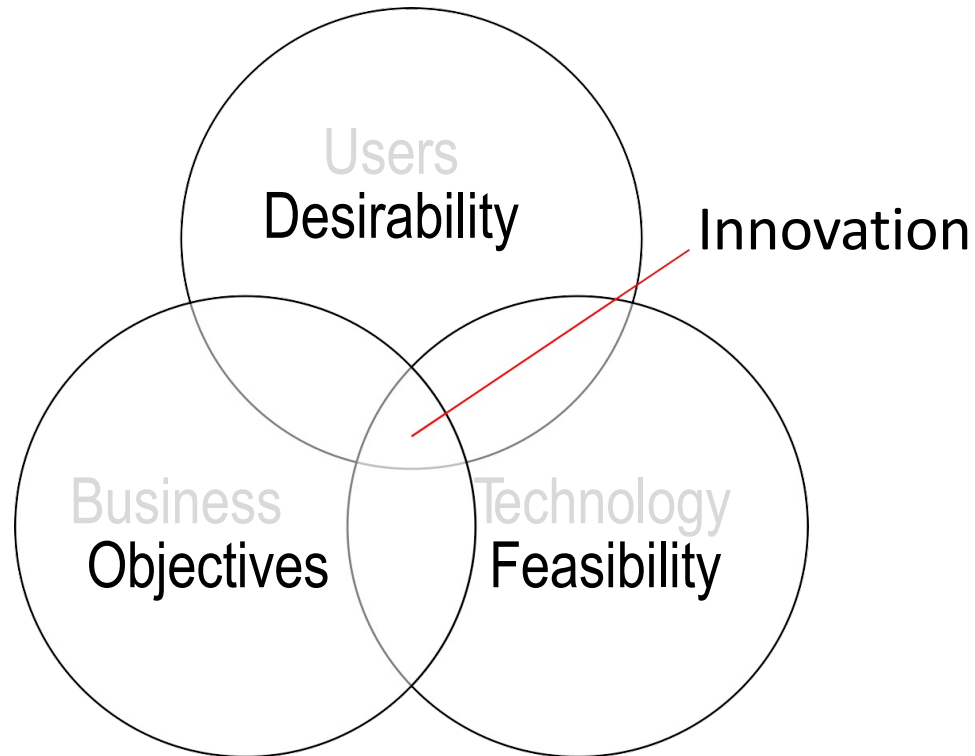


Make

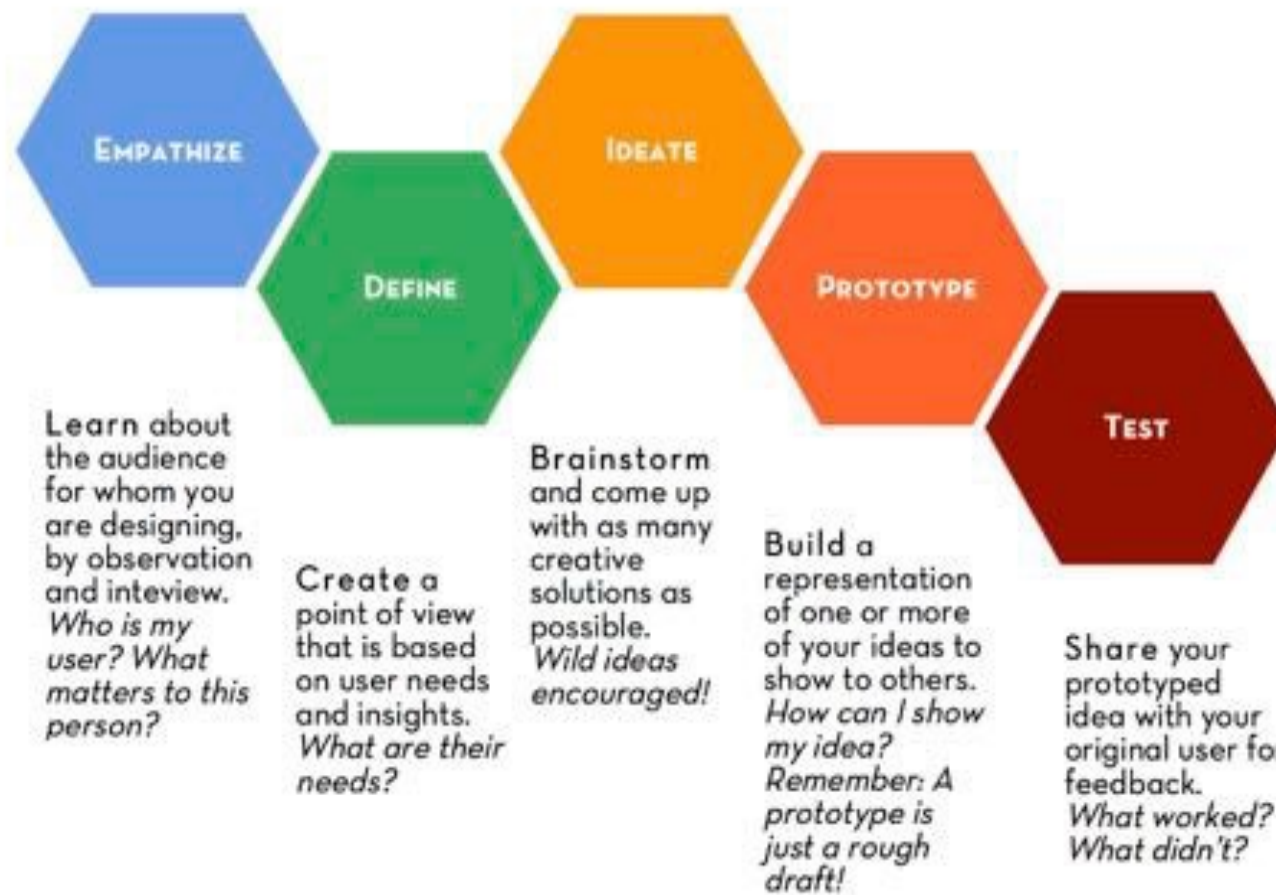
Give concrete form to abstract ideas.

Introduction to design thinking

Trinity for Success



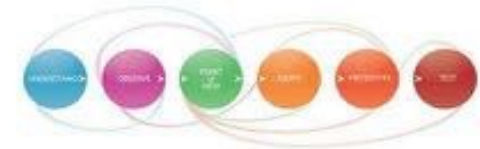
Introduction to design thinking



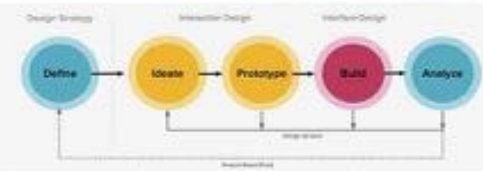
Introduction to design thinking

Design Thinking Models

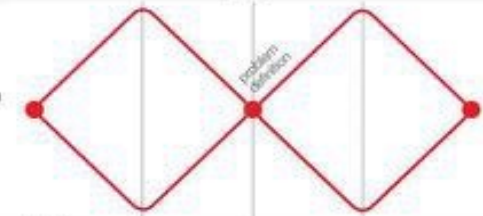
Stanford D.School Design Thinking



Zurb design thinking Model



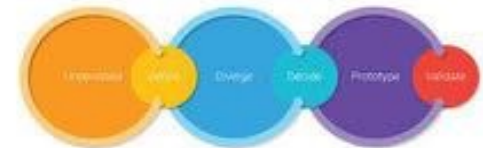
The Double Diamond Diagram by the Design Council



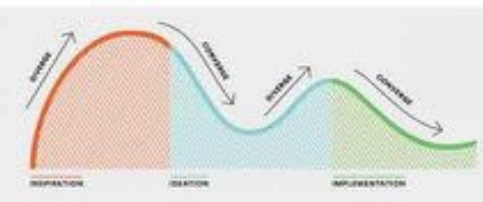
IBM Design Thinking Model



The Google Design Sprint Process



IDEO Human-Centered Design Model



Introduction to design thinking



Course Outcomes

1. Apply design thinking when architecting A.I. solutions
2. Integrate the elements of Design Thinking process
3. Adopt a user-centric approach when designing business solutions
4. Prototype solutions that reduce risks and accelerate results

The Principles

- Keep your user in mind, collaborating with a diverse team, and continuously trying to improve your solutions
- You can't improve your user's as-is experience if you don't understand it clearly. Use design research to learn what your user struggles with today so you can envision the to-be future experience
- We all make assumptions in order to have ideas and make decisions. Just make sure you're testing assumptions with real information

User Research



- Research questions are *not* interview questions.
- Interview and survey questions are the detailed questions you will ask participants in your study; whereas the research questions reflect the high-level purpose of the study.
- A research question can cover several concepts and aspects of the purpose that can then be hashed out in an interview, focus group, or survey.
- Once you have identified a good research question and decided that interviewing people is the best way to answer it, you need to figure out what to ask the people you are interviewing. Interview questions cover topics relevant to the research questions.

Research Question Example

How do families with school-age children decide how to spend money on vacations?

Interview Question Example

Walk me through your last vacation from planning it until when you arrived back home.

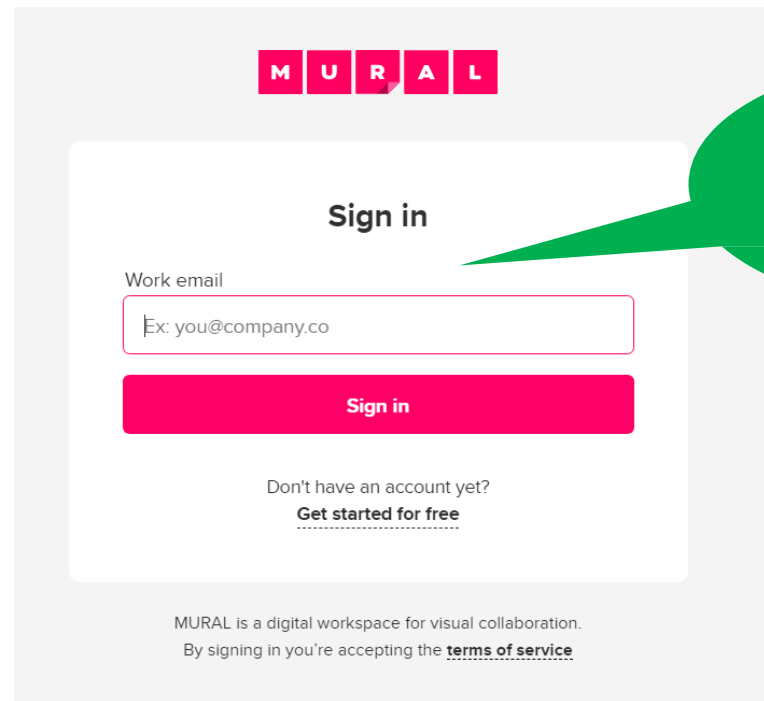
Introduction to MURAL



- Digital workspace for visual collaboration
- Used in IBM Design Thinking Workshops
- Used in AASD4007 for group work & assignments

Introduction to MURAL

MURAL Entry (app.mural.co)

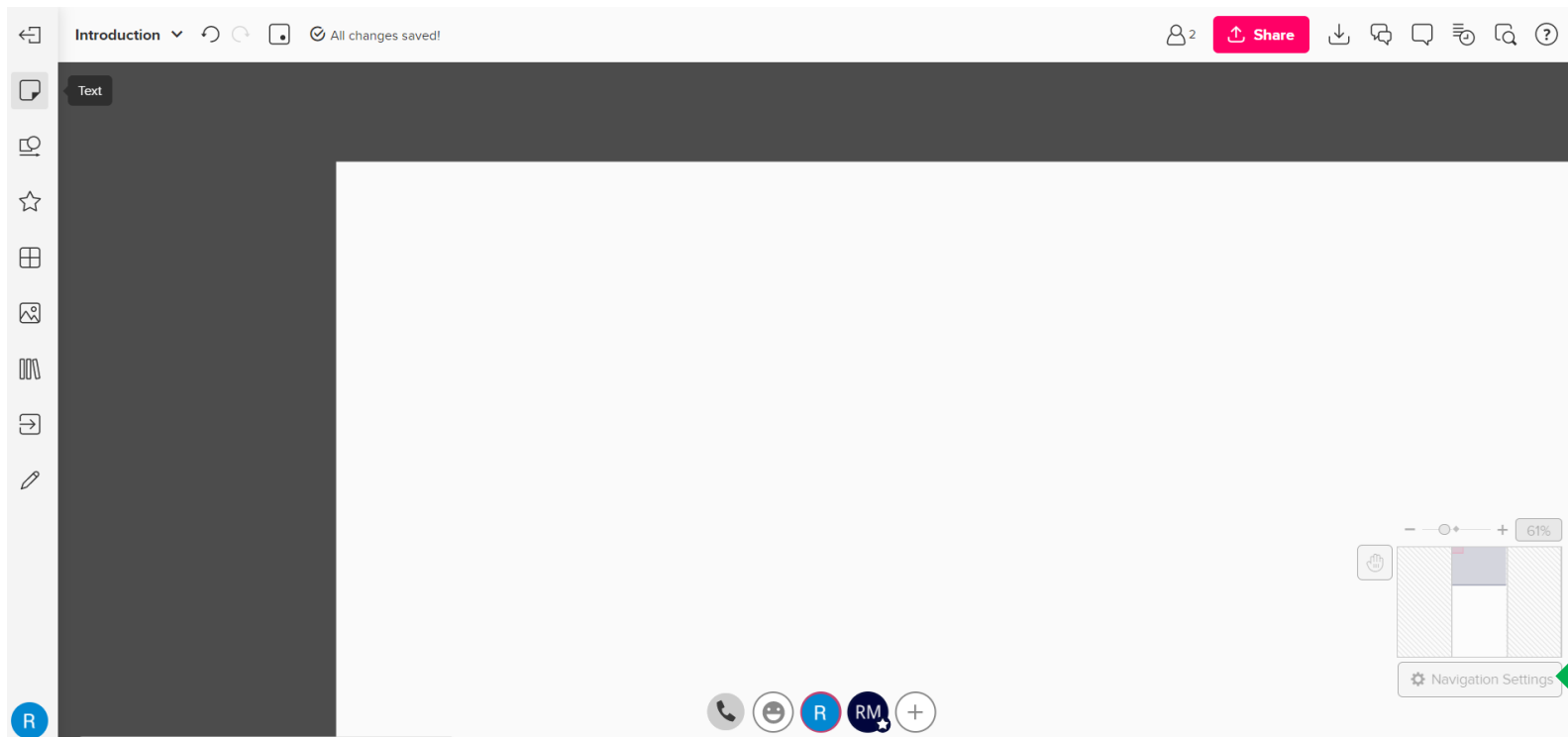


The screenshot shows the MURAL sign-in interface. At the top, the MURAL logo is displayed in pink. Below it, the text "Sign in" is centered. A label "Work email" is positioned above a text input field. The input field contains the placeholder text "Ex: you@company.co". Below the input field is a pink "Sign in" button. Underneath the button, the text "Don't have an account yet?" is followed by a link "Get started for free". At the bottom of the page, a footer states: "MURAL is a digital workspace for visual collaboration. By signing in you're accepting the [terms of service](#)".

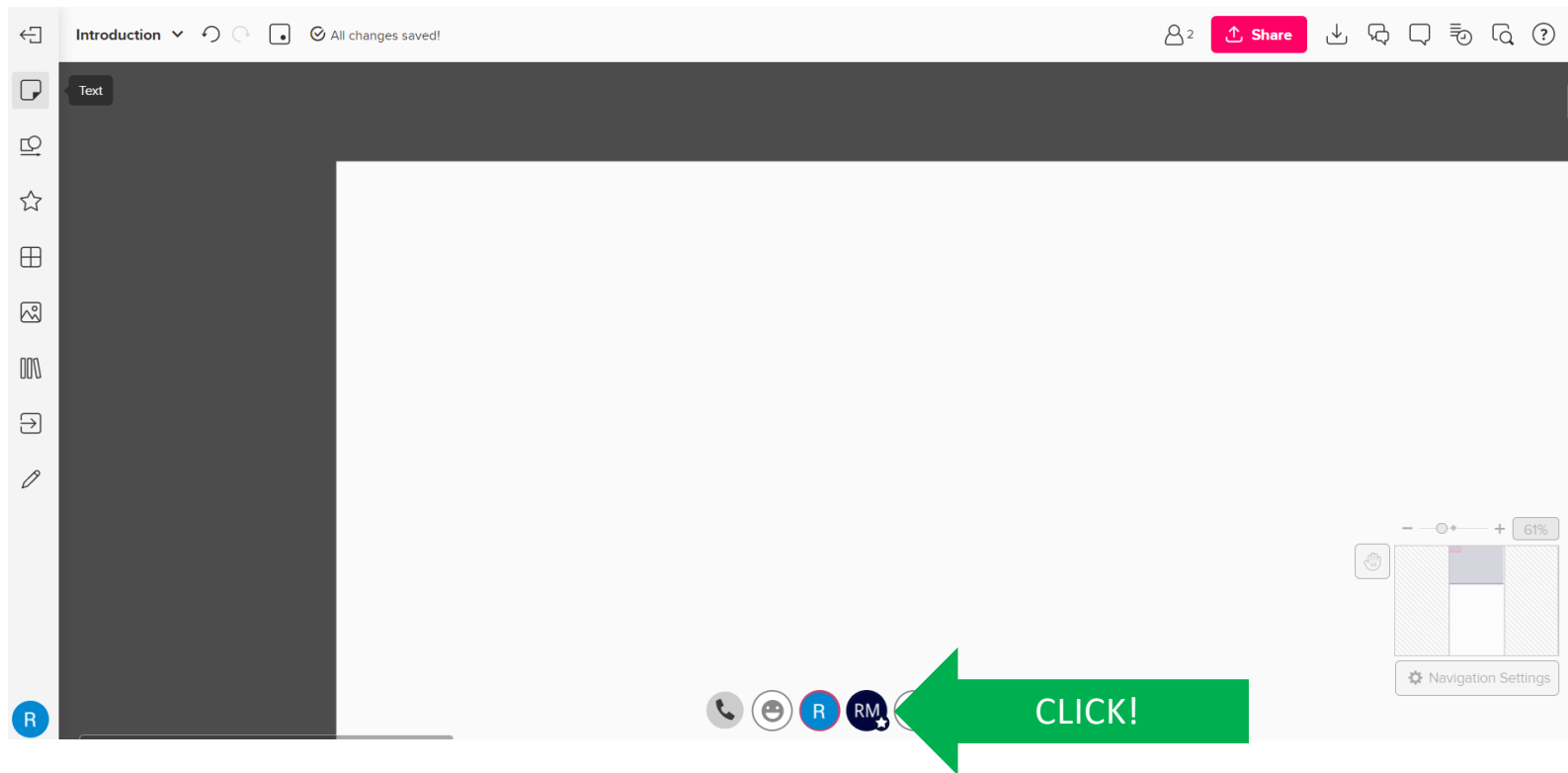
Use your
@georgebrown.ca
email

Introduction to MURAL

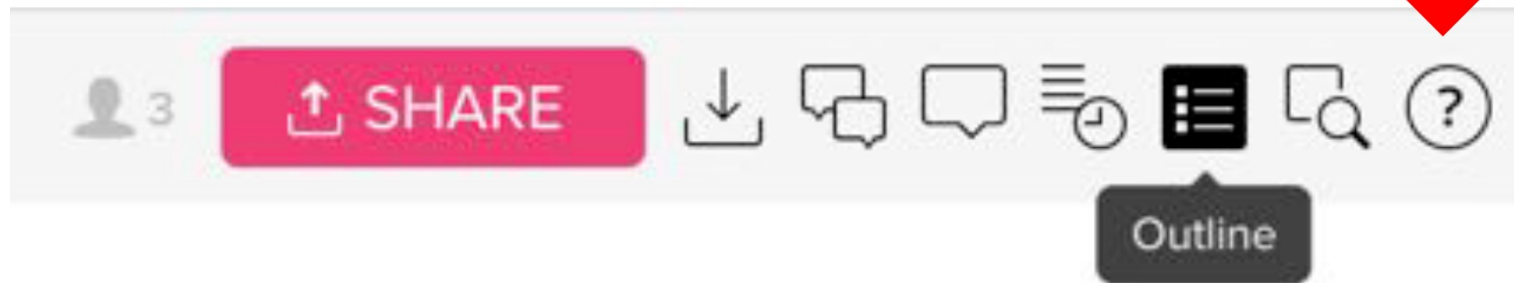
MURAL Entry



Introduction to MURAL

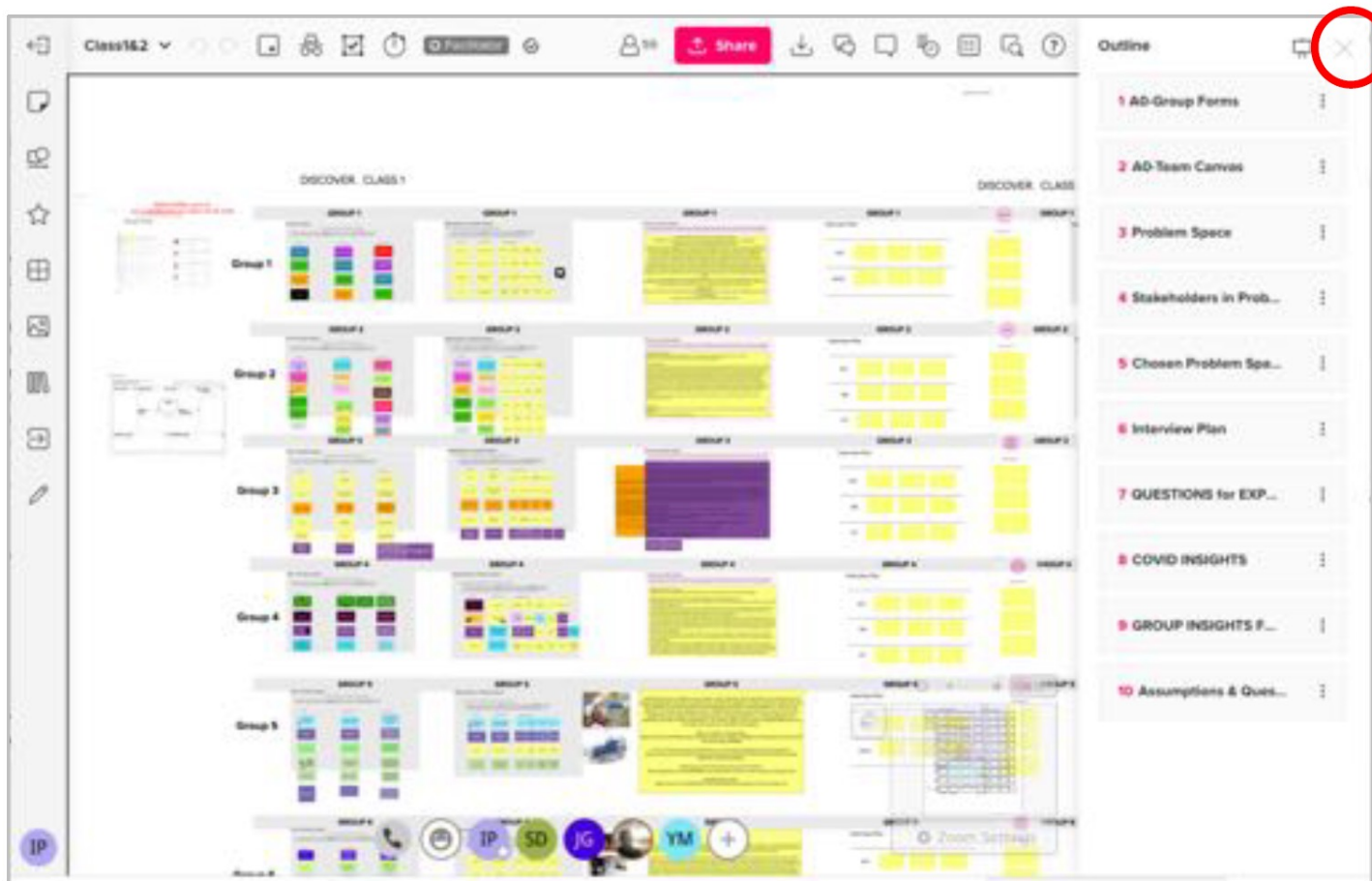


MURAL Tips – Support



support.mural.co/en/

MURAL Tips – Hide Outline



MURAL Tips – Open Outline



MURAL Tips – Share Mural

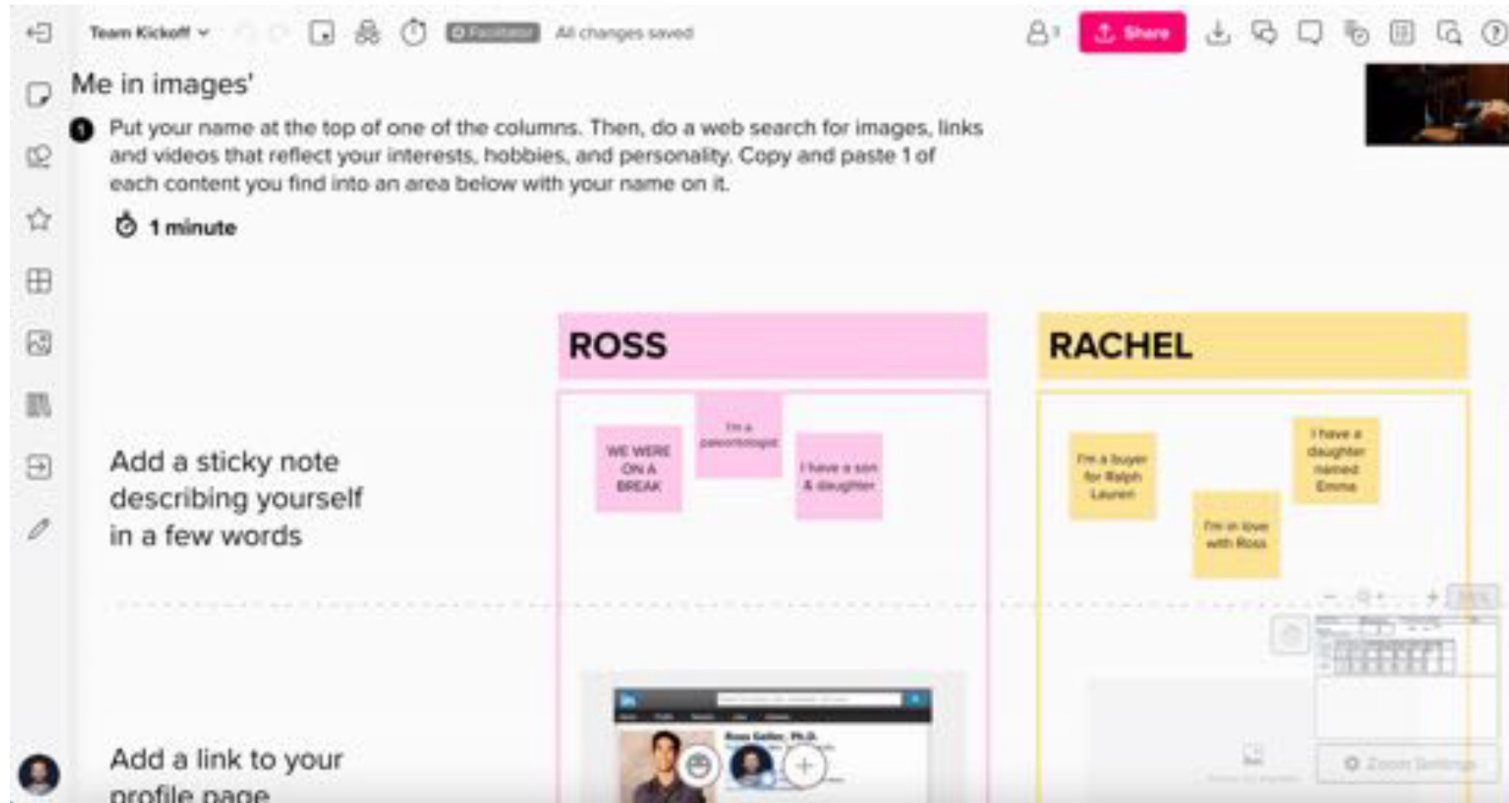
<https://support.mural.co/en/articles/2113777-share-a-mural-and-invite-collaborators>



- **Invite people:** People will automatically join as members of your workspace. Adjust their permissions to 'edit' or 'view'.
- **Visitor Link:** Use this to invite non-members to collaborate quickly and securely.
- **Export:** Get a shareable PDF or PNG of your mural for offline use.
- **Embed:** Generate and copy a code to paste into an external website.

MURAL Tips – Share Link

support.mural.co/en/articles/3354409-share-a-link-to-an-object



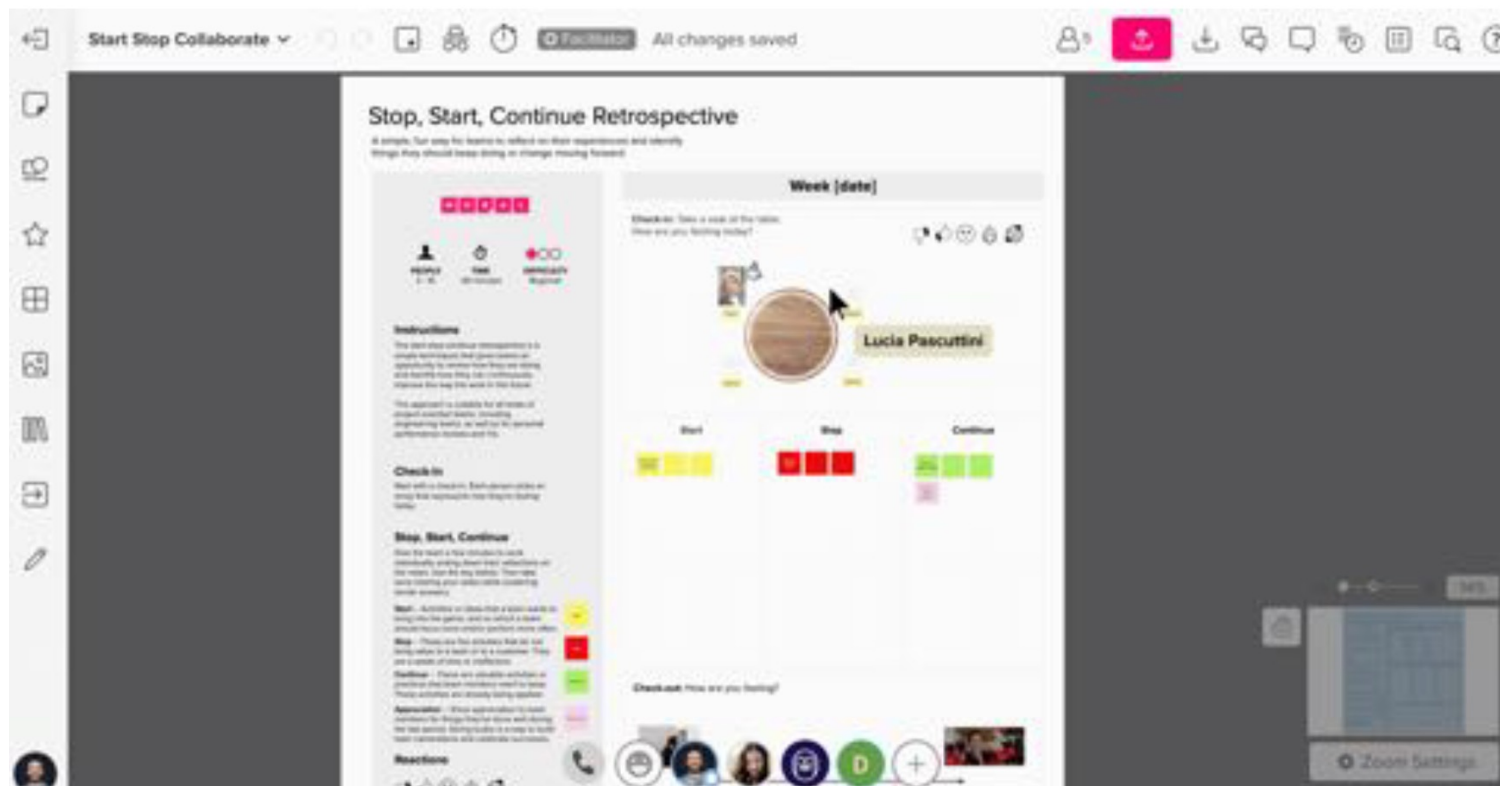
The screenshot shows a Mural workspace titled "Team Kickoff". At the top, there's a navigation bar with icons for various tools and a "Share" button. Below the title, a section titled "Me in images'" contains a numbered instruction: "1 Put your name at the top of one of the columns. Then, do a web search for images, links and videos that reflect your interests, hobbies, and personality. Copy and paste 1 of each content you find into an area below with your name on it." A "1 minute" timer is also visible.

The workspace is divided into two main columns, each with a header: "ROSS" (pink background) and "RACHEL" (yellow background). Below the "ROSS" header, there are three sticky notes: "WE WERE ON A BREAK", "I'm a psychologist", and "I have a son & daughter". Below the "RACHEL" header, there are three sticky notes: "I'm a buyer for Ralph Lauren", "I have a daughter named Emma", and "I'm in love with Ross".

At the bottom of the workspace, there are two sections. On the left, a text prompt says "Add a sticky note describing yourself in a few words". On the right, there's a section titled "Add a link to your profile page" which includes a small image of a person's profile and a "Zoom Settings" button.

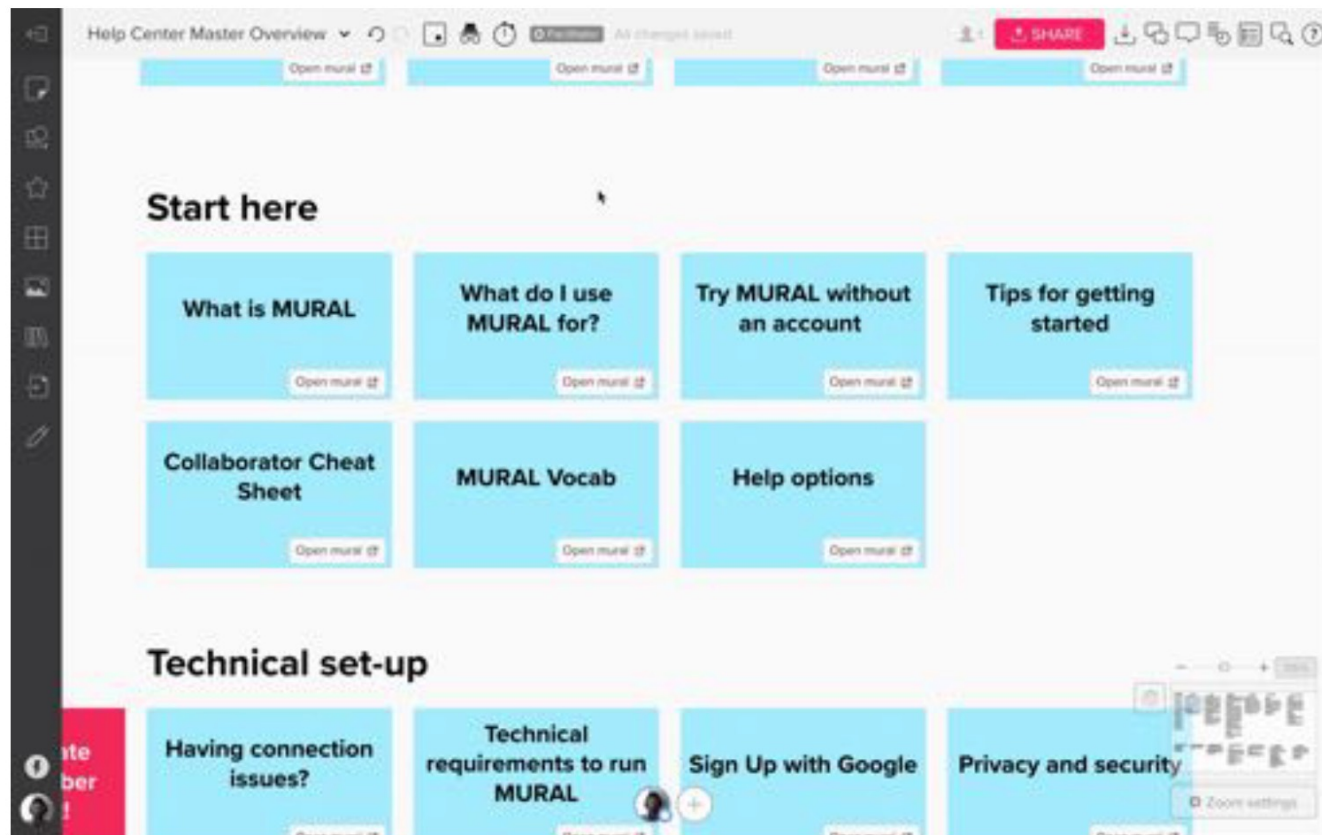
MURAL Tips – Hide Cursors

<https://support.mural.co/en/articles/2113833-hide-cursors>



MURAL Tips – Download Stickies

support.mural.co/en/articles/4507081-download-sticky-notes-text-and-titles-as-a-csv



MURAL Tips – Download Mural



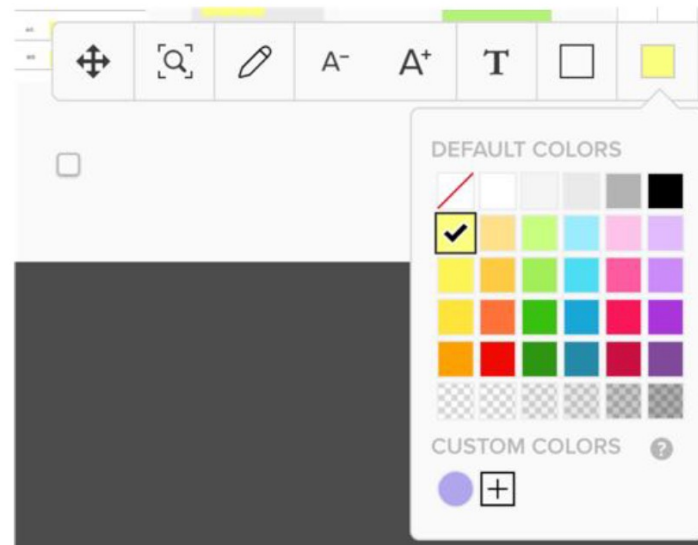
Outline – open & close

Support - Download

Name	Size
 Class1&2 SML.pdf	2.6 MB
 Class1&2-original.pdf	2.6 MB
 Class1&2.png	12.4 MB

MURAL Tips – Colour Your Stickies

- Double click to get a Sticky
- Stickies colours determined by adjacency
- Set your own personalized Stickies colour for group work to help everyone keep track of your contributions



IBM Design Thinking Practitioner



1. Complete the IBM Design Thinking Practitioner online self-study course (approximately 2 hours)
<https://www.ibm.com/design/thinking/page/courses/Practitioner>
2. Post your certificate of completion on BrightSpace.